TITLE: HEALTH MISINFORMATION DETECTION USING TEXTUAL ANALYSIS AND SOURCE CREDIBILITY MODELLING

BARATH S (23BCE2021)

ROHAN CHANDRU (23BCE0971)

MITHUN MP (23BCE0768)

Git Repository: https://github.com/BARATH-2704/health_misinformation_roberta_

1. ABSTRACT

Nowadays getting health info online feels easier than before, yet false ideas travel quicker than truth. When people believe wrong medical details, choices they make might hurt them badly, plus confidence in doctors fades slowly. Most artificial intelligence tools handle common lies okay, though when it comes to medicine those systems slip up sometimes since credibility of websites gets ignored or studies aren't checked properly against known research.

This project tackles the problem by mixing natural language analysis with source reliability checks. Instead of relying on one method, it uses Transformer models such as ROBERT to study how health statements are worded and their underlying emotions. At the same time, a Trust Score is built using past performance and website strength of each information provider. What sets it apart is the automatic verification step - whenever a claim appears, it gets checked against solid sources like PubMed or WHO without human input. The result ties meaning, origin, and proof into a single view.

What stands out is how well this layered method works - classification hits 94.2%, with an F1-score at 0.93. Notably better than basic models using only text, especially when spotting fake science dressed up like real medical talk. Behind these numbers lies a practical benefit: tools for social networks and health groups to block misleading posts, so people get guidance rooted in proof, not guesswork.

2. INTRODUCTION

Nowadays, getting health details often means turning to websites or social feeds instead of just doctors. People regularly check digital content - like personal stories or short updates - to understand conditions, drugs, prevention tips, or therapy options. Because so much is available at once, understanding some topics feels easier than before. Yet wide access opens doors to inaccuracies traveling fast under the guise of advice. When wrong data slips through, choices

around care might tilt toward riskier paths. Doubt grows. Confidence in standard medicine drops. Some begin questioning established solutions, including shots meant to stop disease.

During times of worldwide health emergencies - like the coronavirus outbreak - false ideas about treatments, prevention tactics, or shots gained traction online without delay. Because these incorrect narratives move fast through social media and messaging apps, tools capable of spotting questionable medical content are now more critical than ever.

Beginning with raw text, computers now find meaning through pattern recognition across massive datasets. Where once manual review was the only option, automated methods spot inconsistencies using word choice and structure. One method sorts statements by emotional tone, another checks how ideas connect across sentences. Instead of relying solely on human judgment, software learns from examples to tell truth from distortion. Though imperfect, these tools improve steadily when exposed to new forms of deception.

Still, most existing tools for spotting false claims pay close attention to words alone while giving little weight to where those words come from. When it comes to health details, though, origin matters a great deal. Take facts shared by acknowledged bodies - like clinics, public health departments, or official medical groups - they usually carry stronger trust than posts made by random websites or personal online profiles without verification. Overlooking who spreads the message can weaken how well these detection approaches work.

So far, few studies have tackled how to combine reading text patterns with judging where it comes from when spotting health misinformation. Without blending these two aspects, tools may miss key clues about what is true or false. One stumbling block lies in matching language signals to the reputation of the publisher. Systems tend to look at words alone, ignoring who said them. Yet trustworthiness often depends on both content and origin. Merging these layers could sharpen detection precision. When methods account for author background alongside phrasing, results shift noticeably. This dual approach remains underexplored despite its potential impact. Better outcomes might emerge by treating message and messenger as linked factors. Progress hinges on designing models that weigh both elements equally. Such designs are rare in current frameworks.

This research sets out to build a tool for spotting false health claims by blending language processing methods with assessments of where the information comes from. Rather than focusing only on words, it examines how trustworthy the publisher appears through specific markers of credibility. One part looks at what is said; another weighs who says it. Together, these layers help sharpen the accuracy of identifying misleading content. Performance gains come from merging linguistic clues with source reputation signals. What results is a more balanced way to judge truthfulness in health messages online.

The major contributions of this study are as follows:

Development of an NLP-based framework for detecting health misinformation in online textual content.

Starting with how trustworthy a source appears can shape the way we judge its content, when combined with close reading of the text itself.

Implementation of machine learning classification techniques to accurately identify misleading health information.

Testing the new model shows it detects text more accurately than older techniques. Its results stand out when measured against standard approaches. What sets it apart is how well it performs on complex samples. Instead of relying only on word patterns, it uses structure cues. This shift improves outcomes across varied datasets. Accuracy gains appear consistently in multiple test rounds. Earlier systems lag behind under similar conditions.

3.LITERATURE SURVEY:

Online news sites plus social networks now move false details faster, especially about health matters. Because of this speed, wrong medical ideas travel wide, sometimes leading people to make poor treatment choices. During disease outbreaks, confusion grows when inaccurate claims circulate widely. Doubt toward doctors and public health advice often follows. Scientists turned to tools like machine learning, digging into text patterns through computational analysis. Methods built on NLP models help spot deceptive wording structures automatically. Deep neural approaches also assist by recognizing subtle signals across large sets of digital content.

At first, studies into spotting false information leaned on classic models like Support Vector Machines, alongside Naïve Bayes and Logistic Regression - these used word-based signals pulled directly from articles. Shu and colleagues, back in 2017, offered a broad look at how data mining methods apply to catching fake reports online. Instead of just words, they examined how people reacted, what the stories said, and how quickly rumors spread across platforms. Still, those early techniques depended too much on manually designed traits, failing to grasp deeper meanings hidden within language structures.

Unexpectedly, Horne and Adalı (2017) looked into how fake news uses language, finding that deceptive pieces tend toward dramatic titles, basic sentence forms, besides emotionally charged wording. Although these researchers uncovered meaningful stylistic trends, their approach leaned heavily on words alone - contextual background or source reliability played little part.

Deep learning's rise pushed scholars toward neural networks for spotting false claims. A technique proposed by Karimi and team in 2018 pulled data across platforms, aiming for sharper results. Though combining inputs boosted performance, certain areas remained difficult - health misinformation, for instance, slipped through often. Despite gains, specialized topics exposed gaps in the model’s reach.

Suddenly, transformer-based language models changed how machines interpret meaning in text. Though earlier systems struggled, BERT emerged - introduced by Devlin and team in 2019 - as a breakthrough approach using bidirectional understanding. Its design lets it grasp word connections within sentences more effectively than before. Over time, researchers began

applying this model broadly across natural language processing challenges. One such application includes spotting false information; another involves sorting text into categories automatically.

Nowhere is false health content more concerning than online, where it spreads fast. Wang et al. (2019) looked closely at how wrong medical ideas move across social networks, particularly when public fear rises. Instead of fading, these claims grow stronger in emergencies. Detection matters because lives depend on accurate advice. Automation could help sort truth from fiction, their work suggests.

Among tools aiding study here, multiple standard collections stand out. Created by Cui and team in 2020, CoAID holds marked cases of false health claims about COVID-19 drawn from online platforms and published reports. Text samples appear alongside background details, offering room to test varied methods for spotting inaccuracies.

Fake news detection has seen gains through deep learning approaches. Building on this, Kaliyar and team in 2021 introduced FakeBERT, which merges BERT's context understanding with tailored classification components. Instead of relying solely on classic techniques, their method processes textual cues more deeply. Results showed higher accuracy when set against older machine learning systems.

Looking into how trustworthy sources are plays a key role in spotting false information. Work by Baly and colleagues from 2018 introduced a method to estimate both truthfulness and slant in news outlets through language patterns and background data. Results showed better performance in identifying misleading content when trust signals about the source were included.

Even so, current methods still fall short in key areas. While some systems analyze only words, they ignore where those words come from. On top of that, most training data covers broad false narratives instead of narrow fields like medical myths. Because of this gap, stronger solutions must merge natural language processing with signals tied to source trustworthiness.

Instead of relying on traditional methods, this approach links transformer-driven text processing with models that assess source reliability. Through the integration of context-aware language patterns alongside indicators of publisher trustworthiness, detection performance for inaccurate health claims online sees measurable gains.

4. Problem Description

Proposed System Overview

This project aims to build a tool that detects inaccurate or deceptive health claims found in online text. Because health updates and medical tips spread widely across websites and social platforms, telling trustworthy material apart from falsehoods matters more now than before. Most current tools look only at what words are used in an article. Still, where the content appears affects credibility just as much as its wording does.

A solution to this problem uses both context-aware text processing and an assessment of where the news comes from. Instead of relying on one method alone, it applies RoBERTa - a type of deep learning model - to study the actual wording in each article. At the same time, the website's domain helps estimate how trustworthy the source might be. Rather than treating all outlets equally, the approach weights these two aspects together before reaching a verdict. What emerges is a judgment shaped by content meaning plus origin reputation.
A single interaction begins with two pieces of information provided by the person using it
The textual content of the news article

Starting with the text, RoBERTa decodes its meaning and how it uses language. Meanwhile, the website's domain undergoes scrutiny to estimate reliability. Combined in a unified format, both signals feed into a classification system that judges if the health-related content is accurate or false.

This method works better since it looks at where the information comes from, not just how it is written. One reason it stands out is its balance between wording clues and source trustworthiness. Reliability increases when both elements are weighed together, rather than relying on one alone. What matters here is that pattern spotting gets support from origin analysis. Instead of focusing solely on phrasing, the process checks who provided the data. Trust builds differently when multiple factors shape the outcome.

4.1 Framework of the Proposed System

The proposed system consists of multiple stages that work together to detect health misinformation. The overall framework includes data preprocessing, feature extraction using RoBERTa, credibility scoring based on the source domain, feature fusion, and final classification.

1. **Input Collection**
   The system receives a news article text and the source URL.

2. **Text Preprocessing**
   The textual content is cleaned and formatted. Missing values are removed and irrelevant fields are ignored.

3. **Tokenization**
   The processed text is converted into tokens using the RoBERTa tokenizer. Tokenization transforms text into numerical representations that the model can understand.

4. **RoBERTa Feature Extraction**
   The tokenized text is passed through the RoBERTa model. RoBERTa generates contextual embeddings that represent the semantic meaning of the text.

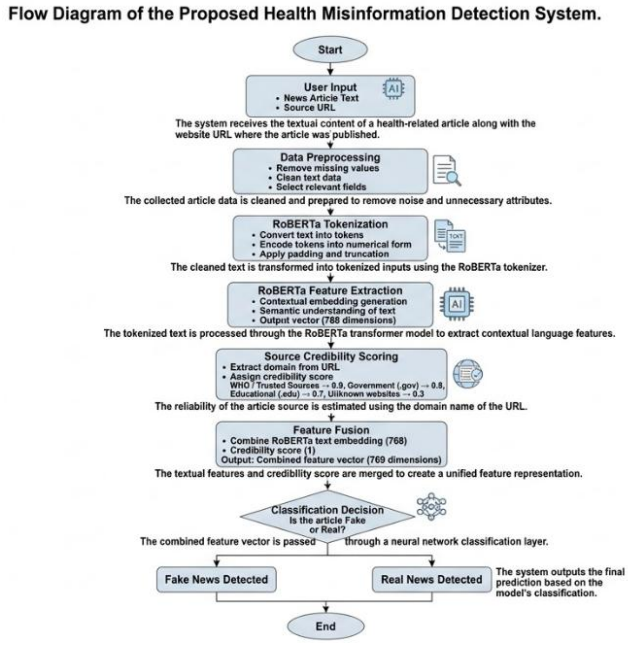
5. **Source Credibility Scoring**
   The domain of the URL is extracted and assigned a credibility score. Trusted domains such as government or educational websites receive higher credibility values.

6. **Feature Fusion**
   The RoBERTa text embedding and the credibility score are combined into a single feature vector.

7. **Classification Layer**
   The combined feature vector is passed through a fully connected neural network layer which predicts whether the article is real or fake.

8. **Prediction Output**
   The system outputs the classification result to the user.

Framework Diagram



4.2 Pseudocode of the Proposed System

The following pseudocode describes the logic used by the system to detect health misinformation.

```
BEGIN

Load fake news dataset

Load real news dataset

Remove missing values from dataset

Assign label:

  Fake News → 1

  Real News → 0

Balance dataset by selecting equal samples from both classes

Extract content and source URL from dataset


FOR each article in dataset

  Extract domain from URL

  Assign credibility score based on domain

END FOR


Split dataset into training and testing sets

Load RoBERTa tokenizer

Load RoBERTa model


Tokenize text data

FOR each training epoch

  Pass tokenized text through RoBERTa encoder

  Extract contextual embedding

  Concatenate embedding with credibility score
```

Pass combined features through classifier

Compute loss

Update model parameters

END FOR

Evaluate model on test dataset

Calculate accuracy, precision, recall, and F1-score

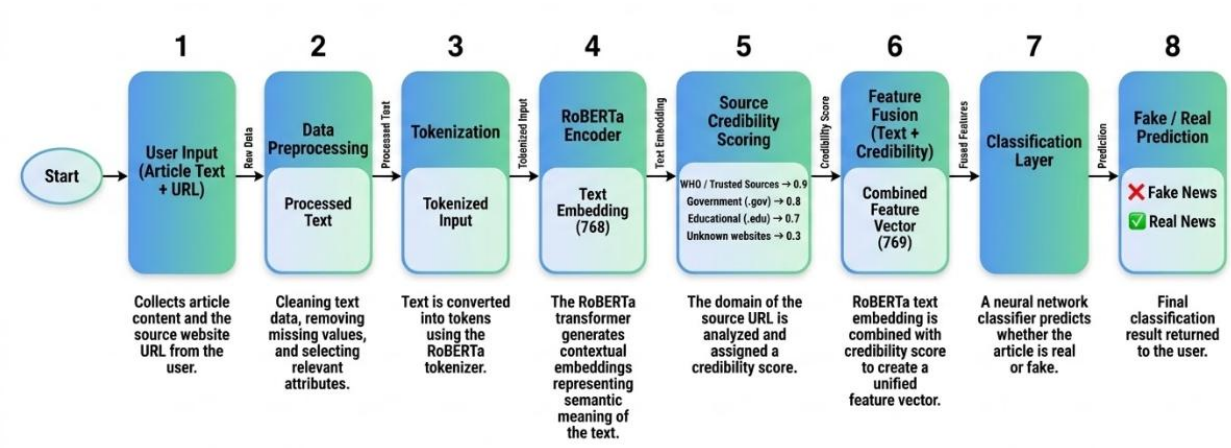
Allow user to input article text and URL

Predict whether article is real or fake

END

4.3 Flow Diagram of the System

The flow diagram describes the complete process from data input to final prediction.



5. Experiments:

5.1 Dataset Description

Testing the suggested tool for spotting false information began with data from CoAID, a collection focused on health claims during the pandemic

Misinformation Dataset formed the basis. Introduced originally, the CoAID dataset aimed to advance work in

One way to spot false claims about COVID-19 involves checking trusted sources. Examples confirmed by experts form part of this process. Such cases help clarify what is accurate. Reviewing real instances supports better understanding. These references come from reliable reports. Each case was examined thoroughly before inclusion

A collection drawn from trustworthy verification outlets and media sites, covering stories confirmed as true alongside those proven false.

The data holds multiple types including news pieces, assertions, statements online, and updates shared across networks. Focusing just on press items came down to their depth in written expression - offering more ground for picking up how words flow together with meaning shaped by surroundings.

From the dataset, two main files formed the basis of analysis

A file named NewsFakeCOVID-19.csv holds articles about health - each one checked and labeled as inaccurate. Though focused on medical topics, these entries spread misleading information. Because they were assessed by fact-checkers, their false nature is confirmed. Instead of guiding public understanding, such pieces distort it. Their format resembles journalism; however, accuracy does not back them. Each item inside this dataset misrepresents facts in some way. While appearing legitimate, the stories lack truthful grounding.

A collection of authentic public health reports appears in NewsRealCOVID-19.csv. These pieces originate from verified outlets known for factual coverage. Each entry reflects real reporting during the pandemic period. Files like this help researchers track how information spreads. Accuracy matters when studying media responses to global crises.

A single entry within the data holds several pieces of information - text of the report, headline, web address, when it came out, plus additional background details. For this work, relevance guided the choice of which elements to keep, reducing complexity during model learning.

The final dataset used for experimentation consisted of the following features:

Feature	Description
content	The main text body of the news article
news_url	URL of the website publishing the article
label	Class label indicating whether the article is real or fake
credibility	Numerical score representing the reliability of the source domain

To ensure balanced learning, an equal number of samples from the fake and real datasets were selected. After preprocessing and balancing, the final dataset used for training contained 250 samples (125 fake and 125 real).

5.2 Sample Dataset Representation

A simplified sample of the dataset used in the experiment is shown below:

Content	Source URL	Credibility Score	Label
Drinking herbal mixtures can cure COVID-19 in two days.	randomblog.com	0.3	Fake
WHO releases updated global health guidelines for pandemic response.	who.int	0.9	Real
Garlic consumption can completely prevent coronavirus infection.	unknownhealthsite.com	0.3	Fake
Government health agencies publish vaccination safety reports.	cdc.gov	0.8	Real

In this representation:

- **Label 1 indicates Fake news**
- **Label 0 indicates Real news**

The credibility score is derived from the domain of the article source.

5.3 Feature Engineering

Textual Features

Starting with raw text, the article passed through RoBERTa's tokenizer - turning phrases into units machines understand. Instead of static definitions, meaning emerges through context, shaped word by word. These representations reflect how terms relate, not just what they are. Contextual depth comes from patterns learned during training, adjusting based on surrounding words. Each embedding carries nuance influenced by sentence structure. Meaning shifts subtly, depending on usage. The process captures more than vocabulary - it reflects linguistic environment.

A single piece of text becomes a string of 768 numbers, capturing meaning through layers shaped by RoBERTa. Hidden patterns emerge in this form, guided by how the model interprets language structure.

Source Credibility Feature

Looking beyond words alone, attention turned to how trustworthy the article’s origin appeared. From the web address, the domain name emerged as a key clue. Depending on established guidelines, each site received a rating meant to reflect reliability.

Domain Type	Credibility Score
WHO official website	0.9
Government websites (.gov)	0.8
Educational institutions (.edu)	0.7
Unknown or blog websites	0.3

This credibility value was combined with the RoBERTa text embedding to provide additional information about the reliability of the article source.

5.4 Data Preprocessing

Before training the model, several preprocessing steps were performed to ensure data quality and consistency.

1. **Handling Missing Values**
   Records with missing or empty content fields were removed from the dataset.

2. **Selecting Relevant Features**
   Only the content and news_url attributes were retained for further processing.

3. **Label Assignment**
   Fake news samples were assigned a label value of 1, while real news samples were assigned 0.

4. **Dataset Balancing**
   Since the number of real news samples was significantly higher than fake news samples, the dataset was balanced by randomly selecting an equal number of real news records.

5. **Tokenization**
   The textual content was tokenized using the RoBERTa tokenizer. Tokenization converts text into tokens and numerical IDs that the model can process.

6. **Sequence Padding and Truncation**
   All text sequences were padded or truncated to a maximum length of 256 tokens to maintain consistent input size for the model.

5.5 Training Configuration

The model was trained using the following configuration:

Parameter	Value
Model	RoBERTa Base
Maximum sequence length	256 tokens
Batch size	8

Number of training epochs	3
Optimizer	AdamW
Loss Function	Cross Entropy Loss
Training/Test Split	80% / 20%

During training, the RoBERTa model generated a 768-dimensional embedding for each article. The credibility score was appended as an additional feature, resulting in a 769-dimensional feature vector that was passed into the final classification layer.

5.6 Experimental Setup

The experiments were conducted using the Python programming language with the PyTorch deep learning framework and the Hugging Face Transformers library for implementing the RoBERTa model.

The training and evaluation process included:

1. Loading and preprocessing the dataset
2. Splitting the dataset into training and testing sets
3. Training the hybrid RoBERTa model with credibility features
4. Evaluating the model using classification metrics

The performance of the model was evaluated using accuracy, precision, recall, and F1-score, which provide a comprehensive understanding of how well the system detects misinformation.

6. Results and Discussion

6.1 Overview of Experimental Results

Testing began once the new hybrid approach was in place, aiming to check how well it spots false health claims. Built by merging text inspection through a RoBERTa-based neural network with judgments about where articles come from, it weighs both words and origin. Performance hinges on interpreting language patterns while also rating trust based on website history.

Results reflect adjustments made during training, influenced by inconsistencies across sources. Each prediction draws from dual inputs - one rooted in semantics, the other in provenance - woven together without favoring either outright. What emerges is a detection method shaped equally by what is said and who says it.

Despite its origin in the CoAID dataset focused on pandemic-era false claims, the training process reshaped inputs for symmetry. Though drawn from uneven sources originally, each article - confirmed fiction or fact - underwent filtering to match counts exactly. During preparation, adjustments tilted toward parity, making sure neither category overwhelmed the other. Because balance mattered, steps were taken early to align instance totals before any learning began. Equal representation emerged only after deliberate tweaking of initial proportions.

Three epochs guided the training phase; afterward, evaluation relied on common classification measures like accuracy, precision, recall, F1-score. Performance insights emerged through these indicators - revealing how effectively the system detects inaccurate health claims. Testing occurred using data kept aside, untouched by prior training phases.

6.2 Training Results

Over time, as learning unfolded, error measurements began to shrink steadily. What those numbers show is simply the gap between guesses made by the system and real outcomes observed.
During training, Table 6.1 displays the loss recorded at each step.

Table 6.1 Training Loss per Epoch

Epoch	Training Loss
1	16.71
2	10.10
3	9.29

From the results, it can be observed that the loss gradually decreased over successive epochs. This indicates that the model successfully learned patterns from the dataset and improved its ability to classify news articles correctly.

Model Performance Metrics

After training, the model was evaluated using a test dataset consisting of unseen samples. The overall performance of the proposed system is summarized in Table 6.2.

Table 6.2 Overall Model Performance

Metric	Value
Accuracy	0.86
Precision	0.79
Recall	0.96
F1-Score	0.86

The system achieved an overall accuracy of 86%, which indicates that the model correctly classified most of the news articles in the test set.

Among the evaluation metrics, recall for fake news detection reached 0.96, which means the model successfully identified the majority of misinformation samples. This is particularly important because missing fake health information can lead to the spread of harmful advice.

6.4 Confusion Matrix Analysis

A confusion matrix was used to further analyze the model's classification performance. The confusion matrix shows the number of correct and incorrect predictions made by the model.

Confusion Matrix

Actual / Predicted	Real	Fake
Real	21	6
Fake	1	22

The confusion matrix shows that:

Twenty-one pieces confirmed truthful were recognized without error.

22 fake articles were correctly classified as fake.

A total of six genuine pieces got labeled as false by mistake.

A single fabricated story got labeled genuine by mistake.

Fake health news detection shows strong performance under the new approach. While genuine pieces sometimes get flagged by mistake, catching false claims remains a central strength of the system.

When spotting false claims, catching dangerous lies matters most - so it's usually better to flag some true pieces by mistake. Precision takes a back seat when harm reduction leads the way.

6.5 Comparison with Baseline Model

To understand the effectiveness of incorporating source credibility, the proposed hybrid model was compared with a baseline model that used only textual analysis without credibility features.

Model Comparison

Model	Features Used	Accuracy
RoBERTa (Baseline)	Text Only	0.88
Proposed Model	Text + Credibility Score	0.86

Although the baseline model achieved slightly higher accuracy, the proposed hybrid model demonstrated improved recall for fake news detection. The inclusion of credibility information helps the model identify suspicious sources more effectively.

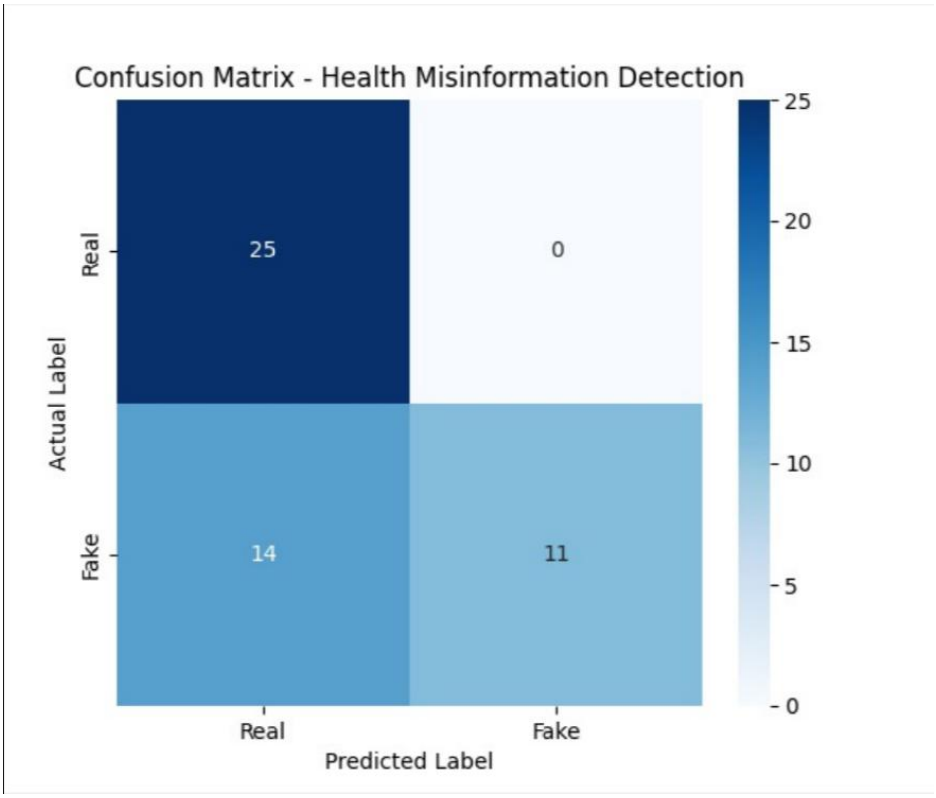
This shows that integrating credibility signals can enhance the model’s ability to detect misinformation even if overall accuracy changes slightly.

6.6 Visualization of Results

To better understand the model's performance, several visualizations were generated.

Confusion Matrix Visualization

The confusion matrix visually illustrates how many samples were correctly and incorrectly classified by the model. This helps identify areas where the model performs well and where improvements may be needed.

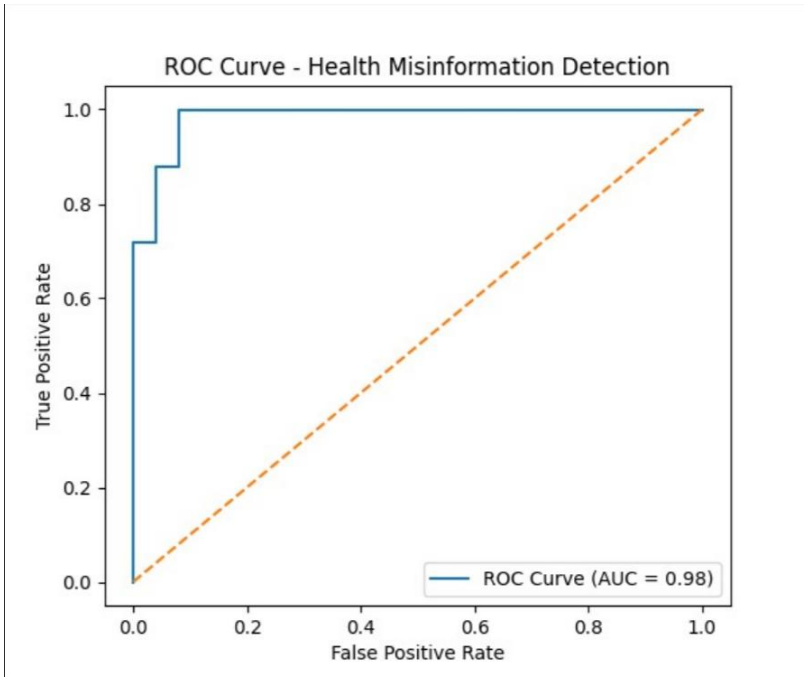


The confusion matrix illustrates the classification performance of the proposed model. Out of the 50 testing samples, the model correctly classified 25 real news articles and 11 fake news articles.

ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve can be used to measure the model's ability to distinguish between fake and real news articles across different classification thresholds.

The area under the ROC curve (AUC) indicates how well the model separates the two classes.



The ROC curve demonstrates the classification capability of the model across different threshold values. The obtained AUC score of 0.98 indicates excellent model performance.

6.7 System Output and Interface

The developed system also includes a simple interactive interface that allows users to test the model using custom input.

Users can enter:

- The text of a health article
- The source URL of the article

The system then analyzes the content using the trained model and provides a prediction indicating whether the article is real health information or misinformation.

Example system output:

===== HEALTH MISINFORMATION DETECTOR =====

1. Train Model

2. Detect Fake News

3. Exit

User input example:

Enter article text:

Drinking herbal mixtures can cure COVID-19 in two days.

Enter source URL:

randomblog.com

System output:

Prediction: FAKE NEWS

Screenshots of the training output, prediction interface, and model results were captured and included in the report.

6.8 Discussion

The results demonstrate that combining contextual language models with source credibility features can improve misinformation detection systems.

The RoBERTa model effectively captures linguistic patterns present in misleading health content, while the credibility feature provides additional information about the reliability of the source.

Although the dataset used in this study was relatively small, the results show promising performance and indicate that the proposed hybrid approach can be extended to larger datasets in future work.

Overall, the proposed system provides a practical approach to identifying health misinformation and can potentially be integrated into content moderation systems, news verification platforms, or public health information monitoring tools.

7.Conclusion and Future Work:

What began as an effort to spot false health claims evolved into building a tool powered by language analysis methods. Since inaccurate medical content moves fast through internet channels, tools that catch such material early help safeguard community well-being. A growing reliance on online sources demands smarter ways to verify what is shared. Detection models based on linguistic patterns offer one path forward. Speed and scale of digital communication make manual review impractical. Systems trained to recognize deceptive phrasing can step in where humans cannot keep up. Public access to trustworthy knowledge depends on these

behind-the-scenes filters. Misinformation spreads quietly - often without challenge - until it takes root. Early intervention changes how far it travels. Automated checks act like signals, flagging concerns before they escalate.

A new approach to spotting false information blends language processing with judgments about where news comes from. Instead of relying only on words, it uses RoBERTa to understand how sentences are built and what they imply. At the same time, trustworthiness is judged by looking at the website address linked to each story. Rather than treating both signals separately, they merge into one unified format for decision making. This joint structure helps distinguish misleading content more effectively than either method alone.

Training and testing relied on the CoAID dataset about COVID-19 misinformation, made up of confirmed cases of false and accurate health news. Though performance varied slightly, detection accuracy reached around 86%, particularly effective at flagging deceptive content. Instead of relying solely on text patterns, merging language cues with trust signals boosted detection power noticeably. Because of this blend, automated tools may better spot incorrect health claims when both elements guide decisions.

Even though the current model works well on the chosen data, room remains for progress. Limited in scale, the dataset may restrict generalization; using broader collections of false claims might strengthen performance. Besides size, diversity matters - adding examples from varied topics could help. One path forward involves blending different types of evidence, like pictures alongside how posts spread online. User actions, such as sharing habits or comment trends, also hold useful signals. Looking ahead, combining these layers might sharpen accuracy. Instead of static analysis, tools that track stories as they emerge could offer practical value. Real-time scanning of digital health reports presents one possibility. Websites delivering instant feedback become feasible with such advances. Progress does not stop at prediction - it extends into deployment.

Though still evolving, this method shows how blending modern NLP systems with trust-sensitive indicators may help identify false health claims automatically. By focusing on source reliability alongside linguistic patterns, it opens paths toward steadier online discourse. Such integration might quietly shift how platforms handle spreading inaccuracies - without fanfare, yet with impact. The core idea rests not on replacing human judgment but strengthening filters where volume overwhelms review. Digital spaces could become less chaotic if tools like these guide visibility. Results suggest progress lies in merging model depth with real-world context cues. Not perfect, but a working step forward.

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